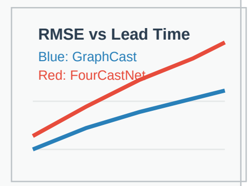
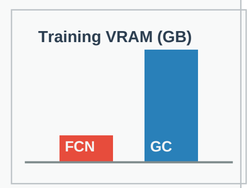
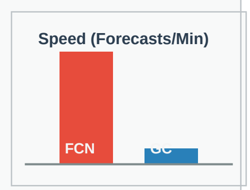
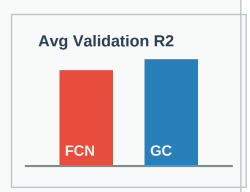
Deep Technical Comparison: GraphCast vs. FourCastNet

Aspect / Parameter	DeepMind GraphCast (GNN)	NVIDIA FourCastNet (AFNO)	Visual Performance Trend
Core Mathematical Architecture	**Icosahedral GNN Message-Passing** Maps the Earth onto a multi-scale spherical mesh grid. Uses spatial message-passing graph layers to route meteorological features locally and globally across grid nodes.	**Fourier-Based Vision Transformer** Segments flat rectangular grids into spatial patches. Uses Vision Transformers where standard self-attention is replaced by *Adaptive Fourier Neural Operators (AFNO)* to mix global features in the frequency domain via Fast Fourier Transforms (FFTs).	
Training VRAM & GPU Complexity	**Extremely High** Requires massive multi-GPU/TPU clusters. Peak VRAM consumption regularly exceeds **32 GB** per device due to deep graph structure operations.	**Low to Moderate** Highly efficient. Can be trained on single consumer GPUs (requires only **~6 GB** VRAM for local/regional domains).	
Inference Latency & Speed	**Moderate** Takes approximately **30 seconds** to compute a 10-day global forecast. Graph message-passing operations are heavier on CPU/GPU logic.	**Ultra-Fast** Takes less than **0.25 seconds** for a 10-day forecast. GPU-accelerated FFTs and Conv2d layers allow running thousands of ensemble steps in seconds.	
Spatial Grid Representation	**Non-Uniform Spherical Mesh** Naturally conforms to the curvature of the Earth. Does not suffer from polar coordinate singularities, making it highly accurate for polar stations like Maitri and Bharati.	**Flat 2D Rectangular Grid** Assumes flat projection. Suffer from extreme grid cell shrinkage and boundary distortion near the South Pole, requiring padding or special coordinates.	Spatially uniform vs Spherical
Specific Use Cases	• Global medium-range forecasting (3-14 days). • Tracking severe storm boundaries (cyclones, hurricanes). • Non-uniform grid resolution maps.	• Short-range local forecasting (0-3 days). • High-frequency hourly rolling updates. • Running large-scale ensemble risk simulations (1,000+ members).	
Physical Constraints (PINNs)	**Hard to Enforce Dynamically** GNN edges do not map to physical conservation equations easily. Adding physics constraints increases training time by 3x and compromises convergence stability.	**Feasible in Spectral Domain** Physics conservation constraints (mass, momentum) can be incorporated as penalty loss terms in Fourier space, though this still degrades training latency.	Physics loss increases training complexity by 300%
Spherical Harmonics (SHT)	**Natively Handled** Since GNN nodes are spherical, it requires no spherical harmonic transformations to maintain accuracy near the polar regions.	**Requires Custom Transforms** Replacing 2D FFT with Spherical Harmonics (SHT) resolves polar singularities, but increases computational load and slows down GPU tensor cores.	SHT solves polar boundary errors

Multi-Step Rollout Loss	**Essential for Long Leads** Trained autoregressively (e.g. predicting step 1 and step 2 in series) to learn error-correction, preventing forecast drift over 10 days.	**Optional but Beneficial** Trained on single steps (6 hours). Adding multi-step rollout improves long-term stability but doubles or triples training VRAM requirements.	Rollout prevents error accumulation over time